

Predictive Maintenance Program Based on Vibration Monitoring

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Abstract. The early detection, identification and correction of machinery problems are paramount to anyone involved in the maintenance of industrial machinery to insure continued, safe and productive operation and to avoid any financial disaster, or personal injury and possible loss of life. Since the vibration analysis can help predict component failure, those parts identified as defective can be scheduled for repair or replacement during planned equipment shutdowns, rather than during costly emergency downtime due to equipment failure. This study presents the use of the machinery vibration as a tool for predicative maintenance program of the rotating machineries. The introduction to the predictive maintenance using vibration analysis in the literature is presented. Causes of vibration and techniques of vibration analysis are illustrated. Construction of the predicative maintenance program via vibration analysis is presented and a computer program named Vibrogram was built up using Visual Basic Language. The Vibrogram is implemented on three rotating machines as case studies at Al- Waha Oil Field, Libya. It was found that the vibration monitoring is a vital component of any good predictive maintenance program for any plant machinery.

Keywords: Vibration, condition monitoring, predictive maintenance, failure.

1 Introduction

The predictive maintenance or condition monitoring is among the planning maintenance part of other maintenances such as a preventive and production maintenances and it involves the trending and analysis of the machinery performance parameters to detect and identify developing problems before a catastrophic failure can occur. It indicates that when replacement of items that wear out should be done so preventing possible major failures. It can detect foreign matters (dirt, oil, moisture) in systems and an appropriate action can be taken [1, 2]. Several parameters can be served as starting point for a condition monitoring. Among these parameters the vibration monitoring for rotating

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machineries, especially for high speed machines such as turbines and compressors. Many industries have found that there are significant advantages of the vibration monitoring including cost saving, reduce the down time and increase the reliability of machines. In addition, the vibration in rotating machineries is not beneficial. It can cause excessive wear, cracking, loosening of fasteners, excessive noise and a host of other problems. In the case of aircraft, serious vibration can cause catastrophic failure leading to loss of life. The goal of vibration monitoring is to detect vibration patterns which will lead to failure.

The vibration monitoring and analysis have been studied by many researchers. The majority of these studies aimed to develop a self-adjusting and integrated monitoring system that is able to function under various working conditions with minimum cost. Hansen and Gao [3] studied the vibration behavior of a deep groove ball bearing with a structurally integrated force sensor. They conducted experimental studies on a ball bearing to validate the analytical and numerical solutions. A good agreement was found between the model-predicted sensor outputs and the experimental results. Mitchell [4], presented a technical article regarding the progress on the vibration measurements and condition based maintenance over the continuous seventy years. Alzadjali and Rameshkumar [5] presented an investigation on the effect of shaft misalignment on the vibration behavior of a centrifugal blower and they observed that the vibration amplitude component at 2X is predominant in all types of misalignment and the vibration increases with the machine speed. In field of wind energy, Sheng and Veers [6] found that the vibration analysis and oil monitoring are the most tools for condition monitoring of wind turbines. Zhun et al [7] developed on line a lubrication monitoring program for a wind mill using a particle filtering technique, in order to reduce the energy cost and improve the wind turbine availability and reliability. In this paper a predictive monitoring program based on the vibration analysis is presented and companion case studies are discussed.

2 Characteristics of Vibration

The vibration is simply a motion of machine or a part of machine back and forth from its equilibrium position. This vibratory motion can be modeled by a motion of mass suspended on end of a spring as explained in the elementary theory of simple harmonic motion [8]. The vibration can be characterized by several common terms. The frequency of the vibration is a number of cycles per second and usually measured in Hertz unit. The amplitude of vibration is a distance that the vibrating part travels back and forth. The increase in amplitude always accompanies an increase of mechanical troubles. The amplitude can be measured by a displacement, a velocity or acceleration. The displacement is a peak to peak displacement and measured in mils, mm or inch. The peak velocity is the rate of change of the displacement with respect to time, measured in mm/s or inch/ s. The peak acceleration is the rate of change of velocity with respect to time and measured in mm/s² or inch/s². Each of the vibration characteristic reveals something significant about the vibration.

3 Common Causes of Vibration

The vibration in rotating machineries occurs due to numerous causes. The unbalance of the rotating shafts is the major cause of vibration in the machines. The vibration may develop due to unequal distribution mass in rotating parts. The points of unbalance produce forces which cause vibration. The unbalance amplitude usually presents at a frequency equal to one time of the operation speed of the part ($1x$). Unbalance may be in single plane (static) or in multiple planes (dynamic unbalance). The second major cause of vibration is the misalignment between shafts of connected machines which generates vibrations that can be mistaken for unbalance. A distinguishing feature is that its principal frequency is at twice running speed ($2x$) and it contains a large axial component. Loose of mechanical parts of the machine generates $1x$ vibration but almost always contain higher harmonics. Worn gears, bent shafts, eccentricity of the bearing, aerodynamic and hydraulic forces, rubbing and resonance are also other sources of vibration in the rotating machineries. Figure 1 illustrates typical examples of the spectral characteristics for some causes of vibrations [9].

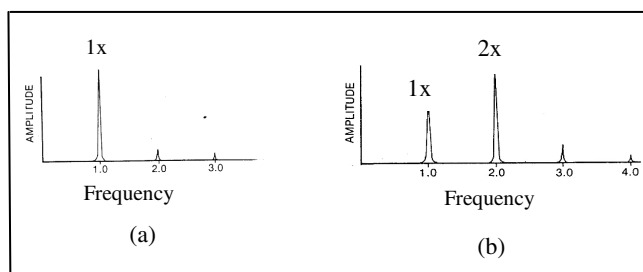


Fig. 1 Typical examples of spectral characteristics for (a): unbalance and (b): Misalignment

4 Analysis Techniques

The causes of vibration are usually identified by several techniques. The choice of the technique depends on the type of the machine and frequency levels. The frequency demand is the common practice to analyze the vibration, where the vibration signals are resolved and displayed as narrow band spectral components. In this technique, the frequency analyzer converts the complex vibration signal from the time domain to frequency domain producing a display of frequency versus amplitude. The other technique is the time domain waveform (orbit analysis). This technique became common practice several years ago when vibration signals displayed on an oscilloscope were utilized for analysis purposes [9]. By establishing the orbit of the rotating shaft on the oscilloscope screen, the picture of the shaft motion within the bearing clearance can be obtain and therefore, the cause of the vibration can be identified. Now days, modern analyzed equipments such as digital and computerized data collectors are used, instead of

the old equipments. These, indeed provide accuracy with rapid analysis to produce an output which may be displayed on the screen of the analyzer immediately.

5 Vibration Monitoring Program

The construction of the vibration program for predictive maintenance of rotating machineries is based on the following factors:

5.1 Plant Survey

The first step in the monitoring program is to study the plant equipments layout, process flow, production patterns, reliability, cost production and maintenance resources. Selection of the machines in the program should be done through the priority of the machines or those are critical to/or in support of, production. When the machines are selected, it should be listed on scheduling sheets.

5.2 Machine Condition

The condition of machines should be studied through the machine maintenance file (if not available, should be constructed). If the mechanical condition of machines is unknown, a complete vibration analysis should be performed. This vibration analysis will detect if machines is trouble free and also will established the start point for monitoring program. Furthermore, vibration levels for machines should be established from the manufacturer, similar machines or expert maintenance persons in the plant. The first vibration checkout may appear machine defects. These defects should be corrected prior to the next measurement, in order to reduce the vibration levels.

5.3 Selection of Measuring Points

The selection of points for vibration measurements should take into account some factors such as machine type, method of mounting, normal direction in which internal forces can be generated, etc. In general, the vibration measurements are usually taken at houses of bearings, because these are locations where rotation parts (shafts) come in touch with the stationary parts (casing). The vibration reading are usually taken in the vertical (V), horizontal (H) and axial (A) directions to the bearing axis of the machine

5.4 Selection of Measuring Instruments

Different modern instruments are available for measuring, analysis and displaying the vibration data. Vibration meters usually hand-held and battery operated are used for measuring displacement, velocity, acceleration and spike energy levels in

a frequency ranges up to 500,000 cycle/min. Data collectors are the instruments which capable of measuring a phase angle, displacement, velocity, acceleration. The modern data collectors are equipped with microprocessor programming capability to measure and store large amounts of vibration data of various parameters. They can measure the displacement, velocity, and acceleration (RMS, pk and pk-pk). Furthermore, collect and store the vibration data on- route and off-route monitoring points and download data to PC for advanced analysis. They can carry out spectrum analysis (FFT) and provide monitoring of rotating parts and dynamic balancing with stroboscope.

Transducers are important instruments, which are used to pick up the vibration readings by converting the mechanical energy into electrical energy. They produce an electrical signal which is function of mechanical vibration. Three types of transducers are commonly used; displacement, velocity and acceleration pickups.

5.5 Period of Vibration Measurements

The length between the check points of the vibration measurements is essential for the monitoring program. The interval of the vibration measurements should be designed according to a number of factors for each check point of each machine. Some of critical machines must be monitored on continuous basis. Periodic, normally monthly, monitoring also can be carried out on the machines according to their speeds. For example, machines operate at low speed (< 500 rpm), the vibration monitoring should be performed every three months, whereas the machines operate between 500 and 3000 rpm, they should be monitoring every month. High speed machines must be monitoring every two weeks [10]. However, other factors may alter the monitoring period such as type of the machines, history of frequent failures, safety and the operation experience which will most likely dictate how often the analysis should be performed. When the vibration interval has been established, it should be fed to the computer as a periodic vibration check chart.

5.6 Vibration Tolerance and Limits

The vibration tolerance should be established for each machine and check point. There are several sources of vibration tolerances are available such as ISO, British and ASME standards. The ISO 10816 Vibration Severity Chart [11] is divided into three zones, the zone A (green) for vibration values from new machines, zone B (yellow) for machines without restriction and zone C (red) for machines in which the damage could occur any time. It also defines four groups of machines, according to their size, base and purpose. In the absent of tolerance information for a machine, the vibration tolerance can be formulated from the maintenance experience or historical data. It should be mentioned that as experience is gained, each machine's tolerance can be reviewed and changed as required. Once the tolerances are established, these are entered on the check round chart.

5.7 *Report of a Vibration Analysis*

Recording system for vibration data should be established in order to issue the vibration report to concerning decision team for planning and scheduling of repairs immediate action. The report is usually a summary of vibration reading of machine trough a certain time (every three months) and it shows the health of the machine in this period of time as the vibration trend is gradually or sudden increase. When repairs have been completed, immediate notification to predictive maintenance team is important. New vibration reading should be taken on the machine after repairing. These will be very helpful in that they will be used to re-evaluate the vibration limits or tolerance for the machine.

5.8 *Training Vibration Staff*

The vibration staff should be having a good knowledge of machinery, trouble shooting and the elementary theory of the mechanical vibration. The vibration analysis requires thoroughness and patience. Those performing the work should be acquainted with the overall objective of the program and what the maintenance department hopes to accomplish. A team effort with vibration engineers, technicians, foremen and operating department heads should be developed.

6 Case Studies

The Waha Oil Field is selected for the applicative part of the study; since the technology of predictive maintenance using vibration monitoring is not fully applied there. The Waha Oil Field is the largest field in the Libyan Waha Oil Company. Three different machines used in three different plants have been chosen for the vibration monitoring program. Table 1 illustrates the technical specifications of each machine. Vibration levels were measured continually with permanently installed on-line vibration measuring systems using velocity or displacement transducers fixed on control panel of the machines. The transducer converts the machine vibration into an equivalent electrical signal that is read on the meter as a vibration level.

Table 1 Technical Specifications of the three machines

Machine	Model	Manufacturer.	Location	Duty	Vib. Alarm	Vib. Trip
GE Gas Turbine	5001	General Electric Co.	Power Plant	Drive a generator	0.5 in /s	1 in /s
Solar Gas Turbine	T-4500	Solar Co.	Gas Plant	Drive a compressor	1.7 mils	2 mils
Centrifugal Pump	DVS-EX	United Co.	Oil Plant	Pump crude oil	1.2 mils	1.8 mils

7 Computer Vibration Program

A vibration program named "**VIBROGRAM**" has been developed using Visual Basic Language, in order to implement the vibration monitoring for the three rotating machineries. Figure 2 shows the flowchart of the program. Once the program has been set up in the computer, the next step is to collect the vibration readings for machines involved. After starting the program, a window will display on the screen for recording the vibration data. A dialogue box will show up asking the technician to follow a few simple instructions to get the task accomplished. As soon as the down load of vibration readings finished, two kinds of reports can be generated; a vibration recording report and /or a graphical report. The reports give an overview about the condition of the machine over a period of data collection. The **VIBROGRAM** records the collection vibration readings from the vibration control panels of the machines for one month period.

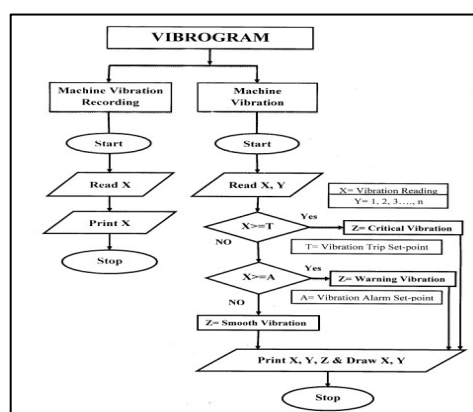


Fig. 2 Flowchart of the computer program "VIBROGRAM" for vibration monitoring

8 Results and Discussion

Table 2 shows the vibration readings of the three machines. The turbines readings were recorded for one month whereas the readings of the pump for fifteen days only, because it was shut down due to production requirements. As it can be seen that the vibration readings of the pump are acceptable and the pump runs in smooth condition.

Table 2 Machines vibration readings at axial direction

GT Turbine (m/s)			Solar Turbine (mils)			United Pump (mils)	
Pedestal Bearing	Comb . Bearing	Turb. Bearing	Comb. Bearing	Turb. Bearing	Pump	Motor Bearing	Pump Support
0.23	0.12	0.15	0.1	0.6	0.4	0.6	0.9
0.27	0.12	0.16	0.1	0.6	0.35	0.5	0.9
0.31	0.13	0.16	0.05	0.8	0.35	0.6	0.9
0.3	0.13	0.16	0.1	1	0.4	0.6	1
0.32	0.15	0.16	0.1	0.7	0.45	0.55	1.1
0.32	0.13	0.15	0.05	0.7	0.4	0.6	0.9
0.32	0.14	0.16	0.1	0.7	0.3	0.65	1
0.33	0.13	0.6	0.2	0.7	--	--	--
0.35	0.14	0.15	0.1	0.7	--	--	--
0.34	0.14	0.15	0.1	0.7	--	--	--
0.34	0.14	0.15	0.2	0.8	--	--	--
0.35	0.14	0.15	0.1	0.8	--	--	--
0.33	0.14	0.16	0.05	0.8	--	--	--
0.33	0.13	0.15	0.05	0.8	--	--	--
0.3	0.14	0.17	0.05	0.9	--	--	--

Figure 3 shows the trend of vibration readings of the GE turbine with time. There are three zones of colors according to the ISO standard [10]: the green zone for smooth region of vibration, the yellow zone for warning region and the red zone for critical region. It can be noted that all the readings are located in the green zone which means that they are within the acceptable range of vibrations. The curve is approximately steady with time. However, the vibration readings increase slightly up to .35 in/s near the warning board. This value is high for such turbo machineries. Therefore, it is recommended that this point should be monitored closely and checked again to analyze the cause of vibration.

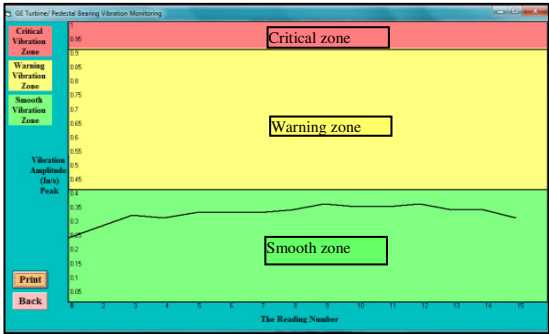


Fig. 3 vibration measurements of the GE turbine

Figure 4 illustrates a window of the Vibrogram monitoring for the bearing of the Solar turbine. Clearly it can be observed that the vibration readings are very low and within the acceptable (smooth) zone. This indicates that the bearing is in good condition.

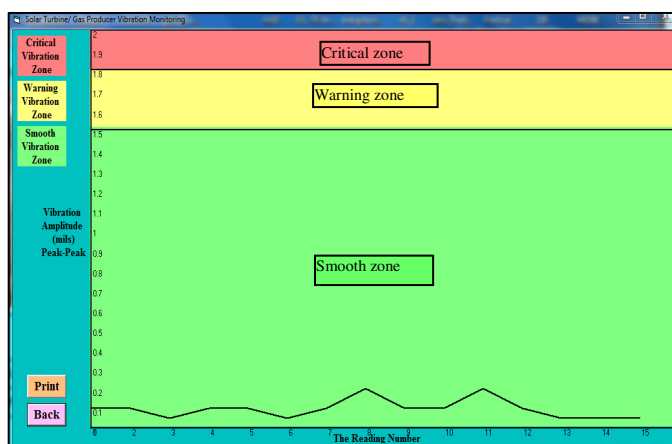


Fig. 4 Vibration measurements of the Solar turbine

9 Conclusions

Outlines of conclusions can be drawn from this study as:

- Vibration analysis provides a powerful tool to locate problems that cause abnormal machine vibration and to prevent catastrophic failure by correcting vibration-related problems.
- Vibration monitoring is a vital component of any good predictive maintenance program for any plant machinery.
- A predictive maintenance program that uses vibration monitoring allows repairs to be scheduled during regular maintenance rather than during costly unscheduled breakdowns, thus yielding significant annual savings in maintenance costs.
- Use of the modern digital instruments can provide an excellent and accurate analysis of vibration readings.

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